

Sajjad Ali Shah

AI Engineer — Data Scientist

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SUMMARY

AI Engineer and Data Scientist with 1+ year of experience designing and deploying Machine Learning, Deep Learning, NLP, Computer Vision, and LLM/RAG solutions. Skilled in Python, FastAPI, predictive modeling, and cloud-based MLOps for building scalable, production-ready AI systems and data pipelines that turn complex data into actionable business insights.

TECHNICAL SKILLS

- **Languages:** Python (Proficient), SQL, Dart, TypeScript, HTML, CSS
- **Machine Learning & Deep Learning:** TensorFlow, Keras, PyTorch, Scikit-learn, CNNs, RNNs, LSTMs, Transformers, Predictive Modeling
- **Generative AI & LLMs:** Hugging Face, OpenAI API, LangChain, RAG, Prompt Engineering, LLM Fine-tuning
- **Data Science & Analytics:** Pandas, NumPy, ETL Pipelines, Power BI, Tableau
- **NLP & Computer Vision:** NLTK, SpaCy, OpenCV, MediaPipe, Matplotlib
- **Backend & APIs:** FastAPI, Flask, REST APIs
- **Cloud, MLOps & DB:** AWS, Docker, Kubernetes, Git, MySQL, Supabase, Pinecone, ChromaDB, FAISS

PROFESSIONAL EXPERIENCE

Harobanx Industries Pvt Ltd — Atomic Energy Pakistan

AI & Software Developer

Islamabad, Pakistan

Mar 2026 – Jun 2026

- Built and deployed the company's official website using Next.js, React, and Tailwind CSS, delivering a responsive, high-performance web platform with modern UI/UX for defense technology showcase and stakeholder engagement.
- Deployed an on-premise RAG chatbot on internal servers using LangChain, self-hosted LLMs, and FAISS to automate support queries offline, ensuring zero data exposure for sensitive defense operations.

Omdena — AI Innovation Challenge: CropLogic

Machine Learning Engineer

USA (Remote)

Sep 2025 – Nov 2025

- Collaborated with a global team of 40+ data scientists across 20+ countries to build predictive models for AI-driven agricultural crop-intelligence solutions.
- Executed data preprocessing and feature engineering in Python; collaborated via Git in an Agile, cross-functional environment to iterate on model performance.

Digital Empowerment Network

Machine Learning Intern

Islamabad, Pakistan

Sep 2024 – Oct 2024

- Trained and fine-tuned supervised ML models (TensorFlow, Scikit-learn), achieving up to 93% accuracy via feature engineering and hyperparameter tuning.
- Built data preprocessing pipelines (Python, Pandas) and validated datasets with cross-functional teams to ensure data quality.

PROJECTS

Saamay AI — Quran Memorization Assistant

Final Year Project

- Designed an end-to-end speech recognition system; fine-tuned speech-to-text (STT) models and deployed real-time, low-latency inference on Hugging Face and AWS.

RAG-Based Medical Chatbot

LLM / Generative AI Application

- Built a Retrieval-Augmented Generation (RAG) system using LangChain and FAISS for medical Q&A, improving response accuracy via semantic search over curated datasets.

Sign Language Detection System

Computer Vision Project

- Developed a real-time ASL recognition system (CNNs, OpenCV, MediaPipe) achieving approximately 90% classification accuracy and integrated it into a cross-platform Flutter mobile app for on-device inference.

EDUCATION

COMSATS University Islamabad (Wah Campus)

BS Software Engineering (CGPA: 3.23/4.0)

Wah, Pakistan

Sep 2022 – Jun 2026

Army Public School & Degree College Malir Cantt

FSc Pre-Engineering

Karachi, Pakistan

Mar 2018 – Aug 2020

CERTIFICATIONS & ACTIVITIES

- **AI Engineering, IBM Data Science, IBM Machine Learning** — Coursera
- **Deep Learning Specialization, Generative AI for Data Scientists, ML Specialization** — Coursera
- **Kaggle Notebooks Master** — Kaggle Data Science Community (2023 – Present)